

Junior Automation & Controls Technician

Daily Study Guide - 7 September 2026 to 28 February 2027

PROGRAMME INFORMATION	
PROGRAMME WINDOW	7 September 2026 - 28 February 2027
WEEKDAY SESSION	3 hours one topic per day
SATURDAY SESSION	4 hours weekly laboratory or active project
SUNDAY	Optional 30-60 minute light review or rest

PURPOSE

The guide and syllabus use the same 25 weeks, topics and project windows. There are no Sunday tests or examination-driven study sessions.

Prepared for Emmanuel Januaria

College-style personal learning programme - not an accredited qualification

How to use this guide

Open the matching syllabus week. Study only the named topic for that weekday. Complete the specific application, record errors as useful feedback and write the next action before stopping.

ONE-TOPIC RULE

If a topic is unfinished after three focused hours, record the stopping point and continue it in the next available study block. Do not compress several unfinished topics into one session.

Three-hour weekday structure

Block	Time	Purpose
Retrieval	15 min	Explain the last relevant idea without scoring yourself.
Focused learning	65 min	Read/watch, define terms and work through the core concept.
Break	10 min	Leave the workspace.
Worked examples	45 min	Calculate, trace, program or interpret with references.
Break	10 min	Reset attention.
Application	25 min	Complete the syllabus problem independently, then correct it.
Error log + next action	10 min	Record misconception, correction, evidence file and next step.

Saturday structures

Week type	Four-hour allocation
Weeks 1-4 laboratory	15 min recall/safety 75 min prepare/build 15 min break 90 min measure/test 15 min break 20 min evidence 10 min error log/next action
Weeks 5-25 active project	20 min choose scope/acceptance criteria 100 min build 15 min break 70 min test/debug 15 min break 20 min evidence and next action

Learning, reflection and evidence cycle

DURING THE WEEK

- ☐ One topic per weekday
- ☐ Brief retrieval without scoring
- ☐ Worked examples with references
- ☐ Independent application
- ☐ Correct errors immediately
- ☐ Save one useful artifact

SUNDAY OPTION

- ☐ Rest is acceptable
- ☐ Review error log if helpful
- ☐ Organize files/evidence
- ☐ Preview Monday briefly
- ☐ No test or score
- ☐ No new topic or submission

Portfolio coordination

Weeks	Saturday mode	Active work
1-4	Weekly laboratory	Only the laboratory matched to that week's electrical topics
5-10	Project focus	Project 1 - Industrial Motor & Conveyor Control System
11-16	Project focus	Project 2 - Automated Process & Instrumentation System
17-25	Project focus	Project 3 - Integrated Remote/Offshore Process Control Simulator

NO EXTRA SATURDAY LABORATORY

When a project window is active, the project itself is the practical work. Do not add a separate generic laboratory. Choose the next incomplete section from your own project plan and use the full four hours on it.

Week 1: 07 Sep - 13 Sep 2026

WEEKLY FOCUS	
Theme	Electrical I
Primary scope	Voltage, current, resistance, Ohm's Law, power and DC circuits
Linked syllabus	M1
Saturday	Not active - use the topic-matched laboratory

Weekly learning outcomes

☐ Explain electrical quantities and units

☐ Rearrange $V=IR$

☐ Calculate $P=VI$

Date	Time	Focus	Specific assignment
Mon 07 Sep	3 h	Voltage	Use the standard 3-hour structure. Complete this application: A battery measures 9 V. State the potential difference and identify positive and negative reference points.
Tue 08 Sep	3 h	Current	Use the standard 3-hour structure. Complete this application: A branch carries 0.025 A. Convert to mA and show where an ammeter belongs.
Wed 09 Sep	3 h	Resistance	Use the standard 3-hour structure. Complete this application: Compare 220 ohm and 1 kohm loads on the same source; predict which draws less current.
Thu 10 Sep	3 h	Ohm's Law	Use the standard 3-hour structure. Complete this application: Calculate voltage for 15 mA through 330 ohms, then calculate current at 9 V.
Fri 11 Sep	3 h	Electrical power and DC circuits	Use the standard 3-hour structure. Complete this application: Calculate power for 9 V at 20 mA and select a supply with reasonable margin.
Sat 12 Sep	4 h	Weekly laboratory	Build a battery-resistor-LED circuit on the breadboard. Predict current, verify polarity, observe normal/open-circuit behavior and record expected versus actual results. Do not short the battery.
Sun 13 Sep	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

☐ What evidence is strongest?

☐ What is the next action?

Week 2: 14 Sep - 20 Sep 2026

WEEKLY FOCUS	
Theme	Electrical II
Primary scope	AC/DC, series and parallel circuits, measurement and multimeter fundamentals
Linked syllabus	M1
Saturday	Not active - use the topic-matched laboratory

Weekly learning outcomes

☐ Distinguish AC from DC

☐ Solve series/parallel networks

☐ Select safe meter methods

Date	Time	Focus	Specific assignment
Mon 14 Sep	3 h	AC and DC fundamentals	Use the standard 3-hour structure. Complete this application: Compare battery DC with mains AC and state why the home trainer remains low voltage.
Tue 15 Sep	3 h	Series circuits	Use the standard 3-hour structure. Complete this application: Calculate total resistance and current for two series resistors on 9 V.
Wed 16 Sep	3 h	Parallel circuits	Use the standard 3-hour structure. Complete this application: Calculate branch and total current for two parallel resistors.
Thu 17 Sep	3 h	Voltage measurement	Use the standard 3-hour structure. Complete this application: Create a safe voltage-measurement plan with reference point and expected range.
Fri 18 Sep	3 h	Current, resistance and continuity measurement	Use the standard 3-hour structure. Complete this application: Choose the correct meter connection for voltage, current, resistance and continuity; identify unsafe energized resistance testing.
Sat 19 Sep	4 h	Weekly laboratory	Build one series and one parallel LED/resistor circuit. Predict node voltages and branch behavior, compare brightness qualitatively, and document one open-branch fault. Use only battery-level voltage.
Sun 20 Sep	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

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☐ What is the next action?

Week 3: 21 Sep - 27 Sep 2026

WEEKLY FOCUS	
Theme	Electrical III
Primary scope	Relays, contactors, circuit protection and industrial control circuits
Linked syllabus	M1
Saturday	Not active - use the topic-matched laboratory

Weekly learning outcomes

☐ Recognize control components

☐ Trace a holding circuit

☐ Explain protection boundaries

Date	Time	Focus	Specific assignment
Mon 21 Sep	3 h	Relays and NO/NC contacts	Use the standard 3-hour structure. Complete this application: Draw coil de-energized and energized contact states.
Tue 22 Sep	3 h	Contactors and auxiliary contacts	Use the standard 3-hour structure. Complete this application: Trace a contactor coil, main contacts and auxiliary feedback without using mains voltage.
Wed 23 Sep	3 h	Fuses, breakers and overload concepts	Use the standard 3-hour structure. Complete this application: Match fuse, breaker and overload functions to faults they address.
Thu 24 Sep	3 h	Start/Stop and seal-in circuits	Use the standard 3-hour structure. Complete this application: Create a Start/Stop truth table with NC Stop and holding contact.
Fri 25 Sep	3 h	Electrical control schematics	Use the standard 3-hour structure. Complete this application: Trace supply-to-load and supply-to-coil paths on a motor-control schematic.
Sat 26 Sep	4 h	Weekly laboratory	Create a breadboard relay/contactor simulation using LEDs as coil, motor and feedback indications. Use jumpers or buttons for Start, Stop and overload; prove seal-in logic with a software or manual state table.
Sun 27 Sep	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

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Week 4: 28 Sep - 04 Oct 2026

WEEKLY FOCUS	
Theme	Electrical IV
Primary scope	Motors, three-phase and VFD fundamentals plus basic troubleshooting
Linked syllabus	M1
Saturday	Not active - use the topic-matched laboratory

Weekly learning outcomes

☐ Explain motor-control layers

☐ Describe VFD interfaces

☐ Troubleshoot systematically

Date	Time	Focus	Specific assignment
Mon 28 Sep	3 h	DC motor fundamentals	Use the standard 3-hour structure. Complete this application: Explain electrical-to-mechanical energy conversion in a small DC motor.
Tue 29 Sep	3 h	Three-phase motor fundamentals	Use the standard 3-hour structure. Complete this application: Interpret a sample three-phase nameplate at awareness level; do not wire mains equipment.
Wed 30 Sep	3 h	Motor protection and overloads	Use the standard 3-hour structure. Complete this application: Separate short-circuit protection from overload protection.
Thu 01 Oct	3 h	VFD fundamentals	Use the standard 3-hour structure. Complete this application: Map VFD ready, run, speed reference, running and fault signals.
Fri 02 Oct	3 h	Basic electrical troubleshooting	Use the standard 3-hour structure. Complete this application: Create a source-to-load troubleshooting sequence for a motor-will-not-run symptom.
Sat 03 Oct	4 h	Weekly laboratory	Simulate a motor/VFD interface using LEDs for Ready, Run and Fault, a jumper for Run command and a state table for speed reference. Inject missing-ready and overload-open faults; record diagnosis and recovery.
Sun 04 Oct	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

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☐ What is the next action?

Week 5: 05 Oct - 11 Oct 2026

WEEKLY FOCUS	
Theme	PLC I
Primary scope	PLC architecture, scan cycle, digital I/O, tags and Ladder basics
Linked syllabus	M2
Saturday	Project 1 active - project focus only

Weekly learning outcomes

☐ Draw PLC architecture

☐ Explain the scan

☐ Write basic rungs

Date	Time	Focus	Specific assignment
Mon 05 Oct	3 h	PLC architecture	Use the standard 3-hour structure. Complete this application: Draw power supply, CPU, DI, DO, communications and field devices.
Tue 06 Oct	3 h	PLC scan cycle	Use the standard 3-hour structure. Complete this application: Explain when an input change becomes visible to cyclic logic and the output update.
Wed 07 Oct	3 h	Digital inputs	Use the standard 3-hour structure. Complete this application: Trace a hobby button or jumper from field state to a PLC-style input tag.
Thu 08 Oct	3 h	Digital outputs	Use the standard 3-hour structure. Complete this application: Trace an output command through interface device to LED or motor representation.
Fri 09 Oct	3 h	Tags and Ladder Logic fundamentals	Use the standard 3-hour structure. Complete this application: Write and test a basic Start/Stop rung with named tags.
Sat 10 Oct	4 h	Project 1 focus	Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 11 Oct	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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Week 6: 12 Oct - 18 Oct 2026

WEEKLY FOCUS	
Theme	PLC II
Primary scope	Core Ladder instructions, latching, timers, counters and one-shots
Linked syllabus	M2
Saturday	Project 1 active - project focus only

Weekly learning outcomes

☐ Read rungs left to right

☐ Use timers/counters

☐ Control repeated events

Date	Time	Focus	Specific assignment
Mon 12 Oct	3 h	Contacts and coils	Use the standard 3-hour structure. Complete this application: Predict rung truth for series and parallel contacts.
Tue 13 Oct	3 h	Seal-in and Set/Reset	Use the standard 3-hour structure. Complete this application: Compare seal-in with Set/Reset and define safe reset behavior.
Wed 14 Oct	3 h	On-delay and off-delay timers	Use the standard 3-hour structure. Complete this application: Create a delayed-start timing chart.
Thu 15 Oct	3 h	Counters and comparisons	Use the standard 3-hour structure. Complete this application: Build a count-to-five state table with reset.
Fri 16 Oct	3 h	One-shots and edge detection	Use the standard 3-hour structure. Complete this application: Explain why a maintained sensor needs one-shot logic for one count per object.
Sat 17 Oct	4 h	Project 1 focus	Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 18 Oct	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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Week 7: 19 Oct - 25 Oct 2026

WEEKLY FOCUS	
Theme	PLC III
Primary scope	Modes, command ownership, permissives and interlocks
Linked syllabus	M2
Saturday	Project 1 active - project focus only

Weekly learning outcomes

☐ Separate permissive/interlock roles

☐ Define mode ownership

☐ Detect command-feedback mismatch

Date	Time	Focus	Specific assignment
Mon 19 Oct	3 h	Permissives	Use the standard 3-hour structure. Complete this application: Write the Boolean conditions required before a motor may start.
Tue 20 Oct	3 h	Interlocks	Use the standard 3-hour structure. Complete this application: Explain how an interlock removes or prevents an unsafe command.
Wed 21 Oct	3 h	Auto/Manual control	Use the standard 3-hour structure. Complete this application: Define what the operator may command in Manual versus Auto.
Thu 22 Oct	3 h	Local/Remote ownership	Use the standard 3-hour structure. Complete this application: Create an ownership table for Local and Remote modes.
Fri 23 Oct	3 h	Command versus feedback	Use the standard 3-hour structure. Complete this application: Design a timer that alarms when RunCmd is true but RunFb remains false.
Sat 24 Oct	4 h	Project 1 focus	Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 25 Oct	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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- ☐ What is the next action?

Week 8: 26 Oct - 01 Nov 2026

WEEKLY FOCUS	
Theme	PLC IV
Primary scope	Sequencing, transitions, controlled restart and recovery
Linked syllabus	M2
Saturday	Project 1 active - project focus only

Weekly learning outcomes

☐ Design ordered sequences

☐ Define transitions

☐ Control recovery

Date	Time	Focus	Specific assignment
Mon 26 Oct	3 h	Sequence requirements	Use the standard 3-hour structure. Complete this application: Convert a conveyor description into numbered steps.
Tue 27 Oct	3 h	Steps and transitions	Use the standard 3-hour structure. Complete this application: Define measurable transition conditions between three steps.
Wed 28 Oct	3 h	State-machine fundamentals	Use the standard 3-hour structure. Complete this application: Draw a state diagram including Idle, Running, Complete and Faulted.
Thu 29 Oct	3 h	Pause, abort and reset	Use the standard 3-hour structure. Complete this application: Specify pause, abort and reset behavior without skipping prerequisites.
Fri 30 Oct	3 h	Power-loss and restart behavior	Use the standard 3-hour structure. Complete this application: Decide what state must occur after power restoration and justify it.
Sat 31 Oct	4 h	Project 1 focus	Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 01 Nov	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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Week 9: 02 Nov - 08 Nov 2026

WEEKLY FOCUS	
Theme	PLC V
Primary scope	Fault handling, alarms, trips, acknowledgement and diagnostics
Linked syllabus	M2
Saturday	Project 1 active - project focus only

Weekly learning outcomes

☐ Design useful diagnostics

☐ Handle faults predictably

☐ Troubleshoot by layer

Date	Time	Focus	Specific assignment
Mon 02 Nov	3 h	Fault detection	Use the standard 3-hour structure. Complete this application: Define five detectable conveyor faults and their evidence.
Tue 03 Nov	3 h	Alarm versus trip	Use the standard 3-hour structure. Complete this application: Classify warning, alarm and trip responses.
Wed 04 Nov	3 h	Acknowledgement versus reset	Use the standard 3-hour structure. Complete this application: Create an acknowledgement/reset state table.
Thu 05 Nov	3 h	First-out and fault memory	Use the standard 3-hour structure. Complete this application: Preserve the first cause when several conditions follow a trip.
Fri 06 Nov	3 h	PLC troubleshooting workflow	Use the standard 3-hour structure. Complete this application: Trace field device, wiring, I/O, logic, HMI and process layers before editing code.
Sat 07 Nov	4 h	Project 1 focus	Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 08 Nov	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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- ☐ What is the next action?

Week 10: 09 Nov - 15 Nov 2026

WEEKLY FOCUS	
Theme	PLC VI
Primary scope	Siemens fundamentals, online diagnostics and Rockwell awareness
Linked syllabus	M2
Saturday	Project 1 active - project focus only

Weekly learning outcomes

☐ Navigate a vendor project

☐ Use diagnostics first

☐ Translate basic vendor concepts

Date	Time	Focus	Specific assignment
Mon 09 Nov	3 h	TIA Portal project structure	Use the standard 3-hour structure. Complete this application: Identify hardware configuration, tags, blocks and watch tables in a TIA project.
Tue 10 Nov	3 h	S7-1200/S7-1500 fundamentals	Use the standard 3-hour structure. Complete this application: Compare compact and modular PLC concepts without claiming hardware experience.
Wed 11 Nov	3 h	Monitoring and diagnostic buffers	Use the standard 3-hour structure. Complete this application: Interpret a simulated diagnostic event before changing logic.
Thu 12 Nov	3 h	Backup and change control	Use the standard 3-hour structure. Complete this application: Create a versioned backup and change note.
Fri 13 Nov	3 h	Allen-Bradley/Rockwell awareness	Use the standard 3-hour structure. Complete this application: Translate Siemens tag/block terminology into introductory Logix concepts.
Sat 14 Nov	4 h	Project 1 focus	Focus on Project 1 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 15 Nov	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
- ☐ What remains uncertain?
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- ☐ What is the next action?

Week 11: 16 Nov - 22 Nov 2026

WEEKLY FOCUS	
Theme	Instrumentation I
Primary scope	Sensors, transmitters, digital/analog signals and 4-20 mA
Linked syllabus	M3
Saturday	Project 2 active - project focus only

Weekly learning outcomes

☐ Trace the measurement chain

☐ Distinguish signal types

☐ Explain live zero

Date	Time	Focus	Specific assignment
Mon 16 Nov	3 h	Sensors and transducers	Use the standard 3-hour structure. Complete this application: Trace a physical variable from sensor to displayed tag.
Tue 17 Nov	3 h	Transmitters	Use the standard 3-hour structure. Complete this application: Explain why a transmitter standardizes the sensor response.
Wed 18 Nov	3 h	Digital instrument signals	Use the standard 3-hour structure. Complete this application: Create a fail-safe level-switch truth table.
Thu 19 Nov	3 h	Analog instrument signals	Use the standard 3-hour structure. Complete this application: Compare 0-10 V and 4-20 mA for noise and fault detection.
Fri 20 Nov	3 h	4-20 mA and live zero	Use the standard 3-hour structure. Complete this application: Calculate percent span at 4, 8, 12, 16 and 20 mA.
Sat 21 Nov	4 h	Project 2 focus	Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 22 Nov	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
- ☐ What remains uncertain?
- ☐ What evidence is strongest?
- ☐ What is the next action?

Week 12: 23 Nov - 29 Nov 2026

WEEKLY FOCUS	
Theme	Instrumentation II
Primary scope	Temperature measurement, selection, scaling and faults
Linked syllabus	M3
Saturday	Project 2 active - project focus only

Weekly learning outcomes

☐ Compare temperature technologies

☐ Scale a signal

☐ Diagnose measurement faults

Date	Time	Focus	Specific assignment
Mon 23 Nov	3 h	Temperature and units	Use the standard 3-hour structure. Complete this application: Convert between degrees C and K and define the measured range.
Tue 24 Nov	3 h	RTD fundamentals	Use the standard 3-hour structure. Complete this application: Explain RTD resistance change and 2/3/4-wire lead effects.
Wed 25 Nov	3 h	Thermocouple fundamentals	Use the standard 3-hour structure. Complete this application: Explain thermocouple millivolts and cold-junction awareness.
Thu 26 Nov	3 h	Temperature transmitters and scaling	Use the standard 3-hour structure. Complete this application: Scale 4-20 mA to 0-100 degrees C.
Fri 27 Nov	3 h	Temperature faults and calibration checks	Use the standard 3-hour structure. Complete this application: Distinguish open sensor, wrong type, wiring error and actual process change.
Sat 28 Nov	4 h	Project 2 focus	Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 29 Nov	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
- ☐ What remains uncertain?
- ☐ What evidence is strongest?
- ☐ What is the next action?

Week 13: 30 Nov - 06 Dec 2026

WEEKLY FOCUS	
Theme	Instrumentation III
Primary scope	Pressure and level measurement
Linked syllabus	M3
Saturday	Project 2 active - project focus only

Weekly learning outcomes

☐ Explain pressure reference

☐ Select level methods

☐ Troubleshoot by cause category

Date	Time	Focus	Specific assignment
Mon 30 Nov	3 h	Pressure fundamentals	Use the standard 3-hour structure. Complete this application: Differentiate gauge, absolute and differential pressure.
Tue 01 Dec	3 h	Pressure transmitter selection	Use the standard 3-hour structure. Complete this application: Select a range and wetted-material assumptions for a small water system.
Wed 02 Dec	3 h	Level measurement technologies	Use the standard 3-hour structure. Complete this application: Compare float, ultrasonic, hydrostatic and simulated level measurement.
Thu 03 Dec	3 h	Level scaling and independent switches	Use the standard 3-hour structure. Complete this application: Scale a level input and define low-low/high-high switches.
Fri 04 Dec	3 h	Pressure/level troubleshooting	Use the standard 3-hour structure. Complete this application: Separate process, installation, wiring, configuration and device causes.
Sat 05 Dec	4 h	Project 2 focus	Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 06 Dec	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
- ☐ What remains uncertain?
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- ☐ What is the next action?

Week 14: 07 Dec - 13 Dec 2026

WEEKLY FOCUS	
Theme	Instrumentation IV
Primary scope	Flow, calibration, valves, actuators and HART awareness
Linked syllabus	M3
Saturday	Project 2 active - project focus only

Weekly learning outcomes

☐ Scale flow signals

☐ Record calibration evidence

☐ Explain final elements and HART

Date	Time	Focus	Specific assignment
Mon 07 Dec	3 h	Flow measurement fundamentals	Use the standard 3-hour structure. Complete this application: Compare pulse, switch and analog flow signals.
Tue 08 Dec	3 h	Flow sensor signals and scaling	Use the standard 3-hour structure. Complete this application: Convert pulse frequency or 4-20 mA into engineering units.
Wed 09 Dec	3 h	Calibration fundamentals	Use the standard 3-hour structure. Complete this application: Prepare three-point as-found/as-left records using simulation.
Thu 10 Dec	3 h	Control valves and actuators	Use the standard 3-hour structure. Complete this application: Define valve command, position feedback, fail position and travel fault.
Fri 11 Dec	3 h	HART fundamentals	Use the standard 3-hour structure. Complete this application: Explain how HART information coexists with a 4-20 mA signal at awareness level.
Sat 12 Dec	4 h	Project 2 focus	Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 13 Dec	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

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☐ What is the next action?

Week 15: 14 Dec - 20 Dec 2026

WEEKLY FOCUS	
Theme	Process Control I
Primary scope	Process variables, PV/SP/MV, feedback and Manual/Automatic control
Linked syllabus	M4
Saturday	Project 2 active - project focus only

Weekly learning outcomes

☐ Label loop variables

☐ Explain feedback

☐ Compare Manual/Auto

Date	Time	Focus	Specific assignment
Mon 14 Dec	3 h	Process variables	Use the standard 3-hour structure. Complete this application: Identify controlled, measured and manipulated variables for a tank.
Tue 15 Dec	3 h	PV and SP	Use the standard 3-hour structure. Complete this application: Calculate SP-PV error and state the assumed control direction.
Wed 16 Dec	3 h	Manipulated variable and disturbances	Use the standard 3-hour structure. Complete this application: Separate controller output from process response and disturbances.
Thu 17 Dec	3 h	Feedback control loops	Use the standard 3-hour structure. Complete this application: Draw process-sensor-controller-actuator feedback.
Fri 18 Dec	3 h	Manual and Automatic control	Use the standard 3-hour structure. Complete this application: Design a bumpless Manual-to-Auto transfer check at fundamentals level.
Sat 19 Dec	4 h	Project 2 focus	Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 20 Dec	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

☐ What evidence is strongest?

☐ What is the next action?

Week 16: 21 Dec - 27 Dec 2026

WEEKLY FOCUS	
Theme	Process Control II
Primary scope	PID fundamentals, alarms, permissives, interlocks and ESD concepts
Linked syllabus	M4
Saturday	Project 2 active - project focus only

Weekly learning outcomes

☐ Explain PID terms

☐ Interpret trends

☐ Design protective responses

Date	Time	Focus	Specific assignment
Mon 21 Dec	3 h	Proportional action	Use the standard 3-hour structure. Complete this application: Predict the directional effect of increasing proportional action.
Tue 22 Dec	3 h	Integral and derivative awareness	Use the standard 3-hour structure. Complete this application: Explain integral removal of offset and why derivative reacts to rate of change.
Wed 23 Dec	3 h	Process response and PID trends	Use the standard 3-hour structure. Complete this application: Classify stable, slow, overshooting, oscillating and saturated trends.
Thu 24 Dec	3 h	Alarm design and deadband	Use the standard 3-hour structure. Complete this application: Create alarm priority, deadband and operator-response entries.
Fri 25 Dec	3 h	Interlocks, permissives, trips and ESD concepts	Use the standard 3-hour structure. Complete this application: Build a cause/effect row for high-high level trip and safe recovery.
Sat 26 Dec	4 h	Project 2 focus	Focus on Project 2 only for 4 hours. Choose the next incomplete project section from your own design/build plan; no separate Saturday laboratory.
Sun 27 Dec	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
- ☐ What remains uncertain?
- ☐ What evidence is strongest?
- ☐ What is the next action?

Week 17: 28 Dec - 03 Jan 2027

WEEKLY FOCUS	
Theme	SCADA/HMI I
Primary scope	HMI/SCADA fundamentals, visualization, hierarchy and PLC communications
Linked syllabus	M5
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Create screen hierarchy

☐ Show state and ownership

☐ Trace communications

Date	Time	Focus	Specific assignment
Mon 28 Dec	3 h	HMI versus SCADA	Use the standard 3-hour structure. Complete this application: Compare local HMI and supervisory SCADA roles.
Tue 29 Dec	3 h	Process visualization principles	Use the standard 3-hour structure. Complete this application: Redesign a cluttered screen using neutral normal states.
Wed 30 Dec	3 h	Screen hierarchy and navigation	Use the standard 3-hour structure. Complete this application: Create overview, area, equipment, alarm and trend navigation.
Thu 31 Dec	3 h	Equipment faceplates	Use the standard 3-hour structure. Complete this application: Design a pump faceplate showing command, mode, feedback, permissives and fault.
Fri 01 Jan	3 h	PLC communications and command ownership	Use the standard 3-hour structure. Complete this application: Trace an HMI command through communications and PLC ownership logic.
Sat 02 Jan	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 03 Jan	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

☐ What evidence is strongest?

☐ What is the next action?

Week 18: 04 Jan - 10 Jan 2027

WEEKLY FOCUS	
Theme	SCADA/HMI II
Primary scope	Alarm management, trends, history, historian concepts and Ignition
Linked syllabus	M5
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Design useful alarms

☐ Interpret trends/history

☐ Navigate Ignition fundamentals

Date	Time	Focus	Specific assignment
Mon 04 Jan	3 h	Alarm philosophy	Use the standard 3-hour structure. Complete this application: Separate alarm condition, priority, message, response and shelving awareness.
Tue 05 Jan	3 h	Alarm priorities and acknowledgement	Use the standard 3-hour structure. Complete this application: Explain acknowledge versus return-to-normal and avoid nuisance alarms.
Wed 06 Jan	3 h	Trends	Use the standard 3-hour structure. Complete this application: Select pens, scales and time windows for a tank diagnostic trend.
Thu 07 Jan	3 h	Historical data and historian concepts	Use the standard 3-hour structure. Complete this application: Distinguish live, historical, stale and bad-quality data.
Fri 08 Jan	3 h	Ignition fundamentals	Use the standard 3-hour structure. Complete this application: Build or plan an Ignition tag, alarm, trend and Perspective view workflow.
Sat 09 Jan	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 10 Jan	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
- ☐ What remains uncertain?
- ☐ What evidence is strongest?
- ☐ What is the next action?

Week 19: 11 Jan - 17 Jan 2027

WEEKLY FOCUS	
Theme	Networking I
Primary scope	Ethernet, topology, IP addressing, subnetting and TCP/IP
Linked syllabus	M6
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Draw network topology

☐ Create an IP plan

☐ Troubleshoot by layer

Date	Time	Focus	Specific assignment
Mon 11 Jan	3 h	Ethernet fundamentals	Use the standard 3-hour structure. Complete this application: Draw PLC, HMI, SCADA and device Ethernet links.
Tue 12 Jan	3 h	Industrial network topology	Use the standard 3-hour structure. Complete this application: Compare a star topology with a small redundant concept at awareness level.
Wed 13 Jan	3 h	IPv4 addressing	Use the standard 3-hour structure. Complete this application: Assign unique addresses and create an IP schedule.
Thu 14 Jan	3 h	Subnet masks and gateways	Use the standard 3-hour structure. Complete this application: Determine whether two /24 addresses are local peers.
Fri 15 Jan	3 h	TCP/IP, ports and ping	Use the standard 3-hour structure. Complete this application: Explain why ping success does not prove an application service is healthy.
Sat 16 Jan	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 17 Jan	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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- ☐ What evidence is strongest?
- ☐ What is the next action?

Week 20: 18 Jan - 24 Jan 2027

WEEKLY FOCUS	
Theme	Networking II
Primary scope	Modbus, OPC UA, serial communications, Profinet and troubleshooting
Linked syllabus	M6
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Explain common protocols

☐ Record serial settings

☐ Diagnose network faults

Date	Time	Focus	Specific assignment
Mon 18 Jan	3 h	Modbus TCP fundamentals	Use the standard 3-hour structure. Complete this application: Map device ID, function, register, datatype and byte order.
Tue 19 Jan	3 h	Modbus RTU and serial settings	Use the standard 3-hour structure. Complete this application: Match baud, parity, stop bits, node address and termination.
Wed 20 Jan	3 h	OPC UA fundamentals	Use the standard 3-hour structure. Complete this application: Explain OPC UA client/server, tags and quality at fundamentals level.
Thu 21 Jan	3 h	Profinet awareness	Use the standard 3-hour structure. Complete this application: Compare Profinet device naming and IP awareness with ordinary Ethernet.
Fri 22 Jan	3 h	Industrial network troubleshooting	Use the standard 3-hour structure. Complete this application: Diagnose link, address, transport, protocol, mapping and application layers in order.
Sat 23 Jan	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 24 Jan	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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- ☐ What evidence is strongest?
- ☐ What is the next action?

Week 21: 25 Jan - 31 Jan 2027

WEEKLY FOCUS	
Theme	Documentation I
Primary scope	P&IDs, schematics, wiring/loop diagrams and registers
Linked syllabus	M7
Saturday	Project 3 active - project focus only

Weekly learning outcomes

- ☐ Trace drawings
- ☐ Coordinate tags
- ☐ Build usable registers

Date	Time	Focus	Specific assignment
Mon 25 Jan	3 h	P&IDs	Use the standard 3-hour structure. Complete this application: Trace equipment, piping, valves and an instrument loop on a tank P&ID.
Tue 26 Jan	3 h	Electrical schematics	Use the standard 3-hour structure. Complete this application: Trace protection, command, coil and feedback on a motor schematic.
Wed 27 Jan	3 h	Wiring diagrams	Use the standard 3-hour structure. Complete this application: Create terminal-to-terminal wiring for a simulated transmitter.
Thu 28 Jan	3 h	Instrument loop diagrams	Use the standard 3-hour structure. Complete this application: Draw process-to-HMI loop boundaries and safe verification points.
Fri 29 Jan	3 h	I/O and instrument lists	Use the standard 3-hour structure. Complete this application: Coordinate tags, ranges, addresses, normal/fail states and descriptions.
Sat 30 Jan	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 31 Jan	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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Week 22: 01 Feb - 07 Feb 2027

WEEKLY FOCUS	
Theme	Documentation II
Primary scope	Narratives, matrices, alarm/network documents and technician reports
Linked syllabus	M7
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Write maintainable documents

☐ Create traceable matrices

☐ Report troubleshooting clearly

Date	Time	Focus	Specific assignment
Mon 01 Feb	3 h	Control narratives	Use the standard 3-hour structure. Complete this application: Write purpose, modes, sequence, protections and recovery in a control narrative.
Tue 02 Feb	3 h	Sequence descriptions	Use the standard 3-hour structure. Complete this application: Define steps and transitions without ambiguous words.
Wed 03 Feb	3 h	Cause/effect and trip/interlock matrices	Use the standard 3-hour structure. Complete this application: Create traceable cause/effect and trip/interlock rows.
Thu 04 Feb	3 h	Alarm lists and network diagrams	Use the standard 3-hour structure. Complete this application: Coordinate alarm tags with a labelled network drawing.
Fri 05 Feb	3 h	Test procedures and troubleshooting reports	Use the standard 3-hour structure. Complete this application: Write expected result, actual result, evidence, defect and retest fields.
Sat 06 Feb	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 07 Feb	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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- ☐ What is the next action?

Week 23: 08 Feb - 14 Feb 2027

WEEKLY FOCUS	
Theme	Safety Awareness
Primary scope	LOTO, permits, electrical hazards, emergency stops and ESD concepts
Linked syllabus	M8
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Identify hazards

☐ Respect authorization limits

☐ Classify protective functions

Date	Time	Focus	Specific assignment
Mon 08 Feb	3 h	Industrial hazard identification	Use the standard 3-hour structure. Complete this application: Identify electrical, stored-energy, rotating, pressure and water/electronics hazards.
Tue 09 Feb	3 h	Lockout/Tagout awareness	Use the standard 3-hour structure. Complete this application: Explain isolation, lock, tag, verification and authorization boundaries.
Wed 10 Feb	3 h	Permit-to-work awareness	Use the standard 3-hour structure. Complete this application: List permit purpose, limits, responsible people and close-out.
Thu 11 Feb	3 h	Electrical safety and test-before-touch	Use the standard 3-hour structure. Complete this application: Create a safe test plan for a low-voltage trainer and state stop-work triggers.
Fri 12 Feb	3 h	Emergency-stop, ESD and hazardous-area awareness	Use the standard 3-hour structure. Complete this application: Differentiate a training E-stop input from a certified safety function.
Sat 13 Feb	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 14 Feb	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

- ☐ Five weekday applications
- ☐ Corrections/error log
- ☐ Saturday lab or project evidence
- ☐ Expected versus actual results

END-OF-WEEK REFLECTION

- ☐ What became clearer?
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- ☐ What is the next action?

Week 24: 15 Feb - 21 Feb 2027

WEEKLY FOCUS	
Theme	Commissioning
Primary scope	Pre-commissioning, I/O checks, loop checks, FAT/SAT and punch lists
Linked syllabus	M8
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Plan commissioning

☐ Execute traceable checks

☐ Control defects and restoration

Date	Time	Focus	Specific assignment
Mon 15 Feb	3 h	Commissioning planning	Use the standard 3-hour structure. Complete this application: Define scope, roles, prerequisites, hold points and safe boundaries.
Tue 16 Feb	3 h	Visual and cold checks	Use the standard 3-hour structure. Complete this application: Inspect labels, polarity, protection, termination and document revision before power.
Wed 17 Feb	3 h	I/O and loop checks	Use the standard 3-hour structure. Complete this application: Prove input stimulus, PLC state, output command, feedback and HMI indication.
Thu 18 Feb	3 h	FAT/SAT-style testing	Use the standard 3-hour structure. Complete this application: Separate factory-style and site-style acceptance evidence for a home simulation.
Fri 19 Feb	3 h	Defects, punch lists and controlled restoration	Use the standard 3-hour structure. Complete this application: Classify, correct, retest and close defects with traceability.
Sat 20 Feb	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 21 Feb	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

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☐ What is the next action?

Week 25: 22 Feb - 28 Feb 2027

WEEKLY FOCUS	
Theme	Integrated Troubleshooting and Handover
Primary scope	Structured fault-finding, regression, demonstration and release
Linked syllabus	M8
Saturday	Project 3 active - project focus only

Weekly learning outcomes

☐ Troubleshoot systematically

☐ Verify correction

☐ Deliver an honest evidence pack

Date	Time	Focus	Specific assignment
Mon 22 Feb	3 h	Symptom-to-root-cause method	Use the standard 3-hour structure. Complete this application: Use symptom, checks, evidence, isolation, diagnosis, correction and verification in order.
Tue 23 Feb	3 h	Electrical and I/O fault isolation	Use the standard 3-hour structure. Complete this application: Trace source, protection, field input, I/O, logic, interface and output.
Wed 24 Feb	3 h	Instrument/process fault isolation	Use the standard 3-hour structure. Complete this application: Separate bad measurement from real process change.
Thu 25 Feb	3 h	HMI/network fault isolation	Use the standard 3-hour structure. Complete this application: Diagnose quality, connectivity, address, protocol, tag mapping and ownership.
Fri 26 Feb	3 h	Regression, handover and technical demonstration	Use the standard 3-hour structure. Complete this application: Run regression, back up files and explain the flagship project in 5-10 minutes.
Sat 27 Feb	4 h	Project 3 focus	Focus on Project 3 only for 4 hours. Choose the next incomplete flagship-project section from your own design/build plan; no separate Saturday laboratory.
Sun 28 Feb	Optional 30-60 min	Light review or rest	No test, score, homework submission or new topic. Optionally review the error log, organize evidence and preview Monday.

EVIDENCE TO KEEP

☐ Five weekday applications

☐ Corrections/error log

☐ Saturday lab or project evidence

☐ Expected versus actual results

END-OF-WEEK REFLECTION

☐ What became clearer?

☐ What remains uncertain?

☐ What evidence is strongest?

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Three-project Saturday roadmap

This table coordinates the study guide with the revised portfolio. It does not prescribe your design; it protects the full Saturday block for your own project decisions.

Project	Dates	Programme weeks	Saturday instruction
Project 1 - Motor & Conveyor	5 Oct-15 Nov 2026	5-10	Focus on Project 1 only for four hours; no separate laboratory.
Project 2 - Process & Instrumentation	16 Nov-27 Dec 2026	11-16	Focus on Project 2 only for four hours; no separate laboratory.
Project 3 - Integrated Flagship	28 Dec 2026-28 Feb 2027	17-25	Focus on Project 3 only for four hours; no separate laboratory.

Project-session record

Field	Record
Chosen scope	_____
Acceptance criteria	_____
Work completed	_____
Expected/actual	_____
Faults/corrections	_____
Evidence saved	_____
Next action	_____

Weekly record - print or duplicate

Record field	Entry
Week and dates	_____
Syllabus module	_____
Five topics completed	_____
Hours planned/completed	_____
Main concept learned	_____
Application evidence	_____
Saturday lab/project result	_____
Error corrected	_____
Evidence location	_____
Next action	_____

RECOVERY RULE

Resume the next unfinished prerequisite. Do not double the following week or convert Sunday into a compulsory catch-up day. Reduce project scope before reducing safety, testing or documentation.

Final readiness checklist

KNOWLEDGE AND PRACTICE

- ☐ Electrical quantities and schematics
- ☐ Motor/control circuits
- ☐ PLC scan, I/O and Ladder
- ☐ Modes/sequences/faults
- ☐ Instrumentation and scaling
- ☐ Process control and PID
- ☐ HMI/SCADA and networking
- ☐ Commissioning/troubleshooting

EVIDENCE AND CONDUCT

- ☐ Exactly three project folders
- ☐ DIY/simulation labels accurate
- ☐ Normal/fault/recovery evidence
- ☐ Drawings/tags/logic agree
- ☐ Test procedures completed
- ☐ Fault reports explain reasoning
- ☐ Backups and revisions controlled
- ☐ 5-10 minute flagship demo

READINESS PRINCIPLE

You are ready to apply when you can explain and demonstrate the work honestly, follow a signal end to end, diagnose faults methodically and show traceable evidence. No Sunday examination is required.